Post-Quantum Signatures in Broadcast Budgets: A Cross-Domain Accounting Study

Author Name
Affiliation

DRAFT – PROVISIONAL SMOKE-TEST NUMBERS

Timings come from smoke runs on a shared cloud VM (`cloud-smoke`, 2026-11-11, commit `3cf20df`; 1000 iterations, 20 for slow schemes). Sizes, scoring and the loss analysis were redone on 2026-11-12 (commits `6ca9403`, `408d1b4`). None are publication numbers. Every value is defined in `numbers.tex` and traced to `results/summary.md`; open items are marked TODO-RESULT or [TODO].

***Abstract*—Handshake-free broadcast channels in cyber-physical systems (drone Remote ID, 5G SIB1, Galileo OSNMA, C-V2X) give each message a fixed number of bytes for authentication. We gather the budgets of these channels from their sources, separating the authentication budget from the payload ceiling. We then account the cost of four authentication modes against them: full signature, signature with certificate handling, a PQ-anchored TESLA-style key chain, and a classical/PQ hybrid. The accounting covers 64 distinct signature size classes from liboqs, plus literature sizes for LMS and XMSS. At the lowest security level of each family, between 5 and 11 of 24 classes fit a single message with a full signature, depending on the profile. All of them are UOV, SNOVA or MAYO, or Falcon under the larger ceilings. On a SIB1 slot model at equal channel use, closed forms show that the chain does not dominate. It verifies at least as many messages as the best fragmentation scheme in 55 of 78 cells, never for ML-DSA-44, while erasure-coded batch signing verifies more but takes much longer. On x86, size rather than timing binds for 7 of 8 families. All numbers are provisional, x86-only and without security analysis.**

***Index Terms*—post-quantum signatures, broadcast authentication, V2X, Remote ID, GNSS, 5G**

I. Introduction

Several cyber-physical broadcast channels authenticate messages that a passive receiver must verify without a handshake: drone Remote ID over Bluetooth pages [1], 5G SIB1 [2], Galileo OSNMA [3] and C-V2X basic safety messages [4], [5]. Each has a hard byte limit per message. Moving them to post-quantum (PQ) signatures, whether standardized (ML-DSA [6], SLH-DSA [7], LMS/XMSS [8]) or selected for standardization (Falcon, as FN-DSA [TODO: cite: FN-DSA standard, none published yet]), is hard because PQ signatures and keys are larger than the classical ones they replace.

Prior work answers this per domain [2]–[5], [9], and within one domain across many algorithms [10], [11]. Studies of different channels use different budget definitions, so their conclusions conflict. For example, Falcon-512 is "insufficient" for C-V2X in [4] and "feasible" in [5]. This short paper is a measurement and accounting study. It proposes no new mechanism, analyses no security properties, and measures no constrained hardware.

Contributions.

- Budget profiles for five channels (seven variants) taken from the sources, each separating the payload ceiling from the authentication budget and marking derived values. The Falcon-512 C-V2X conflict reduces to two different ceilings (Sec. IV-A).
- Mode-level accounting (full, certificate handling, PQ-anchored chain with fragmented anchor, hybrid) for 64 distinct liboqs size classes and 5 LMS/XMSS literature rows. It reports single-message fit, messages needed and time to first authentication.
- A closed-form loss comparison at equal channel use on the SIB1 slot model. It covers the chain, repeated and erasure-coded fragmentation, and Merkle-batch amortized signing, checked under Gilbert–Elliott bursts.
- An x86-only classification of whether size or timing binds, per family, against each profile's packet period.

II. Related Work

Per-domain fit. Twardokus et al. [4] argue that the 437 B C-V2X payload cannot carry a PQ signature and propose a partially hybrid V2V protocol. Abrar et al. [5] find Falcon-512 feasible within an SAE J3161 transport block. Chen et al. [12], [13] study hybrid PQ/ECC certificates for the V2X credential system [14]. Eaton et al. [15] give PQ butterfly key expansion for pseudonym certificates (we read only the abstract). Junquera-Sánchez et al. [3] assess PQ options, including stateful schemes, for OSNMA. Darzi et al. [2] anchor a TESLA-style HMAC chain with one MAYO signature per epoch to fit 5G SIB1. Goertzen and Stebila [9] fragment PQ DNSSEC responses; DNS is request–response, so we keep it only as a ceiling reference. TBRD [16] uses TESLA for Remote ID, and Hussain et al. [17] use a fast MAC for grid sampled values.

Multi-algorithm, one domain. Yoshizawa et al. [10] compare NIST and additional-round signatures for V2X and compress them with zero-knowledge proofs. Mamun et al. [11] review PQ cryptography for ITS. Nabasirye et al. [18] describe a benchmarking framework on Xtensa hardware (abstract only).

Broadcast authentication and amortization. Classical work amortizes one signature over many packets. It includes signing multicast streams over lossy channels [19], signature amortization for multicast [20], tree-based signatures for flows and multicasts [21], and short one-time signatures [22]; we read only titles or abstracts of these. TESLA [23] authenticates with delayed key disclosure, assuming loose

time synchronization and without non-repudiation. Stateful hash-based signatures LMS [24] and XMSS [25] bound the number of signatures per key; few-time schemes [26], [27] trade capacity for speed.

Hybrids, MCUs, envelopes. RFC 9955 [28] defines hybrid signature properties. Fall [29] surveys hybrid deployment and Devevey et al. [30] build compact non-separable hybrids (both abstract only). Cortex-M results exist for ML-DSA [31], MAYO [32], [33], SQIsign [34] and Falcon [35], [36], and pqm4 covers more schemes [37]; all except [31] were read as abstracts. Hyoung et al. [38] build a size–latency envelope for a ROS2/DDS handshake (abstract only), and Sikeridis et al. [39] study PQ signatures in TLS 1.3. Because these were read as abstracts, we cannot exclude an existing broadcast-focused systematization [TODO: check full text of hyoung2026post and fall2025sok].

Maturity. Of the families that fit our budgets, UOV, MAYO and SNOVA, like CROSS and MQOM, are candidates in NIST's additional-signature process, not standards [TODO: cite: NIST additional signatures round status]. Falcon is selected but has no published standard.

III. Method

A. Scheme set

liboqs 0.16.0 [40] enables 221 signature parameter sets, many of which are API variants (e.g. 156 SLH-DSA sets with 6 distinct sizes). We collapse them to 64 classes of distinct (family, NIST level, public key size, signature size). Main tables use the lowest claimed level in each family (24 classes). This is level 1 except ML-DSA, whose lowest is level 2. The signature size S is the *declared* maximum (`length_signature`), not an observed one: Falcon-512 declares 752 B, while our runs observed at most 663 B. LMS and XMSS enter as 5 size-only rows from their RFCs [24], [25], with no timings.

B. Budget profiles

Table I lists each profile's payload ceiling, the bytes available for authentication, and the message period. Remote ID allows 201 B of authentication data (RFC 9575 caps the 255 B field) [1]. SIB1 has 293 B after the 79 B of SIB1 content in the testbed of [2]. For OSNMA the source's block arithmetic is internally inconsistent [3], so we keep two derived variants: 204 B (1632 bit of DSM-PKR crypto material) and 158 B (about 79 bit per block in DSM-KROOT). C-V2X and DNS sources state no authentication-only budget. For these we subtract the overheads the sources state from the ceiling, so their verdicts are upper bounds on fit ("ceiling-based"). Fragment headers are unknown except for DNS (15 B) and are set to 0, so the message counts below are lower bounds.

C. Modes

Let P be the public key size and B the budget. A mode *fits* if its per-message object $x \le B$. Otherwise its cost is $m = \lceil x/B \rceil$ messages and the time to first authentication (TTFA) from the first fragment to the last at the profile's message rate, with no loss. Fragmentation is therefore not a separate mode but the cost of any object that exceeds B.

TABLE I
Budget profiles (bytes). *Derived from stated numbers. ‡Assumed partner. Ceiling-based budgets are upper bounds.

Profile	Ceil.	Auth	Basis	Period	Hybrid	Src.
Remote ID	255	201	auth	3 s	Ed25519	[1]
5G SIB1	372	293	auth	0.16 s	P-256‡	[2]
OSNMA-PKR	208*	204*	auth	480 s	P-256	[3]
OSNMA-KROOT	208*	158*	auth	480 s	P-256	[3]
C-V2X 437	437	373*	ceiling	0.1 s	P-256	[4]
C-V2X J3161	2481	2414*	ceiling	0.1 s	P-256	[5]
DNS	1232	1232	ceiling	n/a	P-256	[9]

- *full*: $x = S$; the key is provisioned out of band.
- *full_cert*: the certificate handling of the profile's source. C-V2X sends an 8 B digest on four of five messages and the full certificate on every 5th. Remote ID sends a 16 B DET and the key via a separate endorsement. SIB1 and OSNMA deliver the key outside the scored message. DNS sends signature plus key in the first response. A full certificate is $S + P + f$ with $f = 105$ B of metadata and issuer signature, taken from one Falcon-512 data point in [5] and applied to every scheme. The mode fits only if both the digest message and the certificate message fit.
- *chain*: per message a 16 B tag [2] and a 16 B disclosed key (32 B). Once per epoch an anchor of $S + 20$ B (16 B commitment, 4 B timestamp) is fragmented over as many messages as needed. Key, commitment and timestamp sizes are our assumptions. The join time is the anchor's TTFA plus one message period of key delay.
- *hybrid*: $x = S + 64$ B, with the classical scheme the profile's source uses. This is Ed25519 for Remote ID and ECDSA-P256 (raw encoding; DER takes up to 72 B) elsewhere. The SIB1 partner is assumed.

D. Loss model (Q2)

We use only the SIB1 slot model: $C = 293$ B of authentication per slot and one slot per 160 ms [2]. Every scheme gets the same M slots per message, so channel use is equal. A signature needs $k = \lceil S/C \rceil$ fragments. Let p be the i.i.d. slot loss and $q = 1 - p$. *frag_rep* repeats all k fragments $r = \lfloor M/k \rfloor$ times, so the verified share is $(1-p^r)^k$. *frag_mds* erasure-codes M fragments, so the share is $\Pr[\mathrm{Bin}(M, q) \ge k]$. *chain* sends t copies of the tag message and reserves the rest of the window for the anchor, so the share is $(1 - p^t)(1 - p^{tw})(1 - p^{r_a})^{k_a}$, where k_a and r_a are the anchor's fragments and repetitions over an epoch of E messages. *merkle_rep/mds* sign a batch of n messages once and spend the spare slots on the batch signature. Messages that never verify count as missing; TTA95 is the 95th-percentile time to authenticate over all messages. Assumed parameters are $M \in \{3, 6, 12, 24, 48\}$, $E = 20$, at least 2 anchor repetitions, $w = 2$ key windows and $n \in \{4, 8, 16\}$. A slot-level simulation (20 seeds $\times$ 4000 messages) agrees with the closed forms within 0.0121 (3 of 240 cells beyond 3 standard

TABLE II
LOWEST-LEVEL LIBOQS CLASSES FITTING ONE MESSAGE (OF 24). §CHAIN COUNTS THE 32 B STEADY-STATE TRAFFIC ONLY; THE ANCHOR IS FRAGMENTED AND COSTED SEPARATELY.

Mode	RID	SIB1	PKR	KROOT	CV437	J3161	DNS
full	7	8	7	5	8	11	11
full_cert	0	8	7	5	0	4	0
hybrid	5	7	5	0	8	11	11
chain§	24	24	24	24	24	24	24

TABLE III
SHARE VERIFIED / TTA95 IN SLOTS (160 MS EACH) AT $M = 12$. "MISS": FEWER THAN 95% EVER VERIFY. k AT $C = 293$ B: FALCON-512 3, MAYO-2 1, ML-DSA-44 9.

Sig.	p	chain	frag_mds	frag_rep	merkle_mds	merkle_rep
Falcon-512	0.1	1.000/14.2	1.000/5	1.000/6	1.000/252	1.000/264
	0.3	0.998/15.3	1.000/7	0.976/12	1.000/276	0.993/336
MAYO-2	0.1	1.000/14.2	1.000/2	1.000/2	1.000/216	1.000/216
	0.3	1.000/15.3	1.000/3	1.000/3	1.000/228	1.000/228
ML-DSA-44	0.1	0.914/miss	0.974/12	0.387/miss	1.000/264	0.991/354
	0.3	0.428/miss	0.493/miss	0.040/miss	1.000/294	0.782/miss

errors). We use it only as a check and for Gilbert–Elliott bursts (mean burst 2, 5, 10 slots, same stationary loss).

E. Size versus timing (Q3)

For each (lowest-level class, profile) pair, excluding DNS (no message rate), the pair is *size-bound* if $S > B$. It is *timing-bound* if the larger of the p99 sign and verify times exceeds the profile's packet period: 0.1 s for C-V2X, 0.16 s for SIB1, 3 s for Remote ID and 30 s per OSNMA block. For classes with only 20 iterations no tail is estimated; they count as timing misses only if the median already exceeds the period. A family is size-bound if size-only pairs outnumber timing-only pairs. Keygen and receiver verification throughput are out of scope.

IV. RESULTS

A. Q1: what fits a single message

Table II and Fig. 1 give the fits. A full signature fits one message in 5 to 11 of 24 classes, depending on the profile. Under the authentication budgets (Remote ID, SIB1, OSNMA) every fitting class is UOV, SNOVA or MAYO-2; Falcon fits only under the C-V2X J3161 and DNS ceilings. ML-DSA-44 (2420 B) fits no profile, not even the 2414 B J3161 budget. CROSS, MQOM and SLH-DSA fit none. Among the stateful rows, only LMS (1292 B, derived, 32 signatures per key) fits, and only J3161.

Cost when it does not fit. ML-DSA-44 needs 9 SIB1 messages (TTFA 1.28 s), 7 C-V2X messages (0.6 s) and 13 Remote ID messages (36 s). On OSNMA, one block per 30 s makes any multi-message object slow: Falcon-512 takes 4 messages and 1740 s on the PKR variant.

Certificates. Certificate handling fits only where the key travels outside the scored message (SIB1, OSNMA) or under the J3161 ceiling (4 classes). Remote ID, the C-V2X 437 B profile and DNS fit no class, because keys of the small-signature families are large: 43 576 to 412 160 B for UOV and 4912 B for MAYO-2. All *full* rows assume the key is pre-provisioned.

Chain and hybrid. The chain's 32 B per message fits all 168 (class, profile) pairs; its cost moves to the anchor and the join time. For ML-DSA-44 the join takes 1.44 s on SIB1, but on OSNMA-PKR it takes 6210 s (960 s for MAYO-2). The 64 B hybrid partner removes fits wherever the PQ signature is near the budget, and all of them on OSNMA-KROOT.

The C-V2X Falcon-512 conflict. Twardokus et al. use the 437 B maximum payload of a 10 MHz channel [4]. Abrar et al. assume a J3161 20 MHz profile at MCS 11 with 10 sub-channels, giving a 2,481 B transport block [5]. With the stated overheads subtracted (budgets 373 B and 2414 B), Falcon-512 (752 B declared) needs 3 messages under the first ceiling and fits one under the second, even with a full certificate (1754 B peak). Both papers are thus consistent with their own ceilings. "Feasible" in [5] means fitting a transport block; their own 90% delivery target is met only in the lightest line-of-sight scenario.

Baselines. From stated crypto sizes and the 293 B budget we reproduce the 13 and 34 SIB1 packets of [2]. For OSNMA we compute 68 DSM blocks at the source's 79 bit per block, where it reports 67 [3]. The remaining baselines are not reproduced (Sec. V).

B. Q2: chain vs. fragmentation vs. batching at equal channel use

With channel use equal, no scheme dominates (Fig. 2, Table III). Where fragmentation can run at the message rate, the chain verifies at least as many messages as the best fragmentation scheme in 55 of 78 (signature, M, loss) cells, and at least as many as every scheme in 54. Per signature, the counts are Falcon-512 30/30, MAYO-2 25/30 and ML-DSA-44 0/18. The chain beats fragmentation outright only where fragmentation cannot run at the message rate (ML-DSA-44 at $M < k$). Even there, batch signing verifies more messages in most cells. Erasure-coded batch signing verifies the most messages (1.000 for ML-DSA-44 at 30% loss), but its TTA95 in Table III is 216–294 slots. The chain needs about 14–15 slots and fragmentation 2–12. Erasure-coded fragmentation is the fastest whenever the signature fits a few slots. The earlier simulation in which fragmentation beat the chain at low loss gave fragments two repetitions and chain messages none; at equal channel use that effect disappears.

Burst loss lowers the share verified for the chain and the erasure-coded schemes. At 30% stationary loss with mean bursts of 10 slots, the chain drops from 0.998 to 0.874, frag_mds from 1.000 to 0.853, and merkle_mds from 0.989 to 0.882. Averaged over the burst cells, the drop is 0.013 for the chain and 0.025 for frag_mds. Repetition schemes change by -0.051 and -0.014; a negative value means bursts helped.

C. Q3: size or timing

Size binds in 7 of 8 families and timing in 0 (Table IV); UOV fits everywhere and meets every period. The "both"

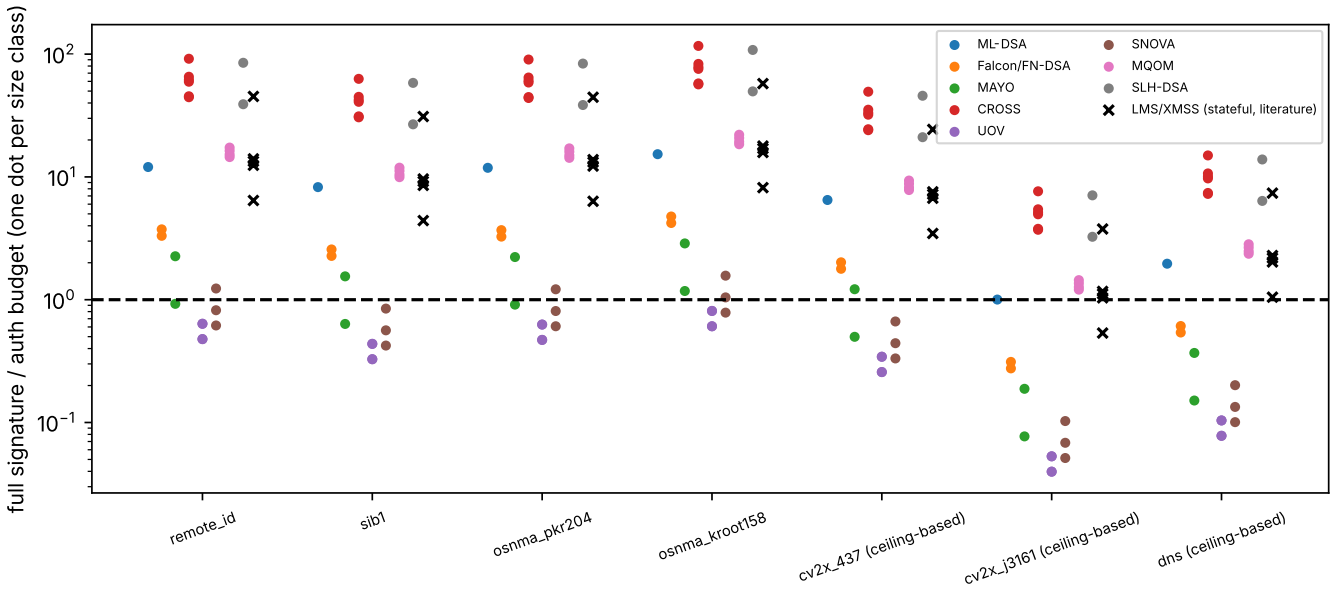


Fig. 1. Full signature (declared size) divided by authentication budget, one dot per lowest-level size class; crosses are LMS/XMSS literature sizes. Below the dashed line fits one message. Provisional.

TABLE IV
(CLASS, PROFILE) PAIRS BY BINDING CONSTRAINT, X86 ONLY. NO PAIR WAS UNDECIDABLE. †TIMING FROM 20 SAMPLES (MEDIANS). N.D.: NOT DISTINGUISHING.

Family	size-only	timing-only	both	neither	binds
CROSS†	36	0	0	0	size
Falcon	10	0	0	2	size
MAYO	6	0	0	6	size
ML-DSA	6	0	0	0	size
MQOM†	24	0	0	0	size
SLH-DSA†	9	0	3	0	size
SNOVA	4	0	0	14	size
UOV	0	0	0	24	n.d.

pairs are SLH-DSA, whose 128s set has a median sign time of 486 ms. This result is close to tautological. The packet periods (0.1 s to 30 s) are long compared with x86 operation times, such as a p99 sign time of 253.3 μs for ML-DSA-44 and 250.3 μs for Falcon-512. It says nothing about microcontrollers. Against the 3 ms grid deadline quoted from the IEC 62351-6 text in [17], which belongs to none of our profiles, 0 of 6 lowest-level UOV variants miss. So do 2 of 12 UOV variants over all levels, both compressed-key/secret-key variants with alternatives at the same level.

V. LIMITATIONS

- **Provisional numbers.** The timings come from one smoke run on a shared cloud VM (CPU: TODO-RESULT) with 1000 iterations (20 for CROSS, MQOM and SLH-DSA, whose tails are not estimable). Sizes and timings use one 64 B message, so per-message variance is not sampled. Publication runs (10 000/200 iterations) are pending.
- **x86 only.** No MCU timings were measured [TODO: cite or measure: Cortex-M sign/verify for UOV, SNOVA, MAYO]; Q3 does not cover constrained devices.
- **No security analysis.** The modes differ in trust model. The chain gives no non-repudiation and needs loose time synchronization [2], [23]. Hybrids differ in how separable their components are [28]. We compare bytes, not security, and do not model chain continuity across epochs.
- **Keys pre-provisioned.** *full*, *chain* and *hybrid* assume the public key is already at the receiver. The larger keys (UOV, MAYO-2, SNOVA) cannot be sent in any single message.
- **Certificate overhead.** The $f = 105$ B certificate overhead is a single Falcon-512 data point applied to all schemes. C-V2X certificate metadata (TODO-RESULT) is not counted, and mixed PQ-CA/classical-subscriber chains are not modelled.
- **Budgets.** The C-V2X and DNS budgets are ceiling-based upper bounds. The OSNMA budgets are derived from inconsistent source arithmetic. Fragment headers for Remote ID, SIB1, OSNMA and C-V2X (TODO-RESULT, TODO-RESULT, TODO-RESULT, TODO-RESULT) are unknown and set to 0. DNS has no message rate (TODO-RESULT) and is not a broadcast channel.
- **Loss model scope.** Q2 uses only the SIB1 slot model, three signatures (Falcon-512 and MAYO-2 at level 1, ML-DSA-44 at level 2), assumed chain and batch parameters, and synthetic loss. There are no measured traces and no cold-start or join analysis.
- **Baselines not reproduced.** These are the 324/744 B OTA overhead (TODO-RESULT) and 10%-loss point (TODO-

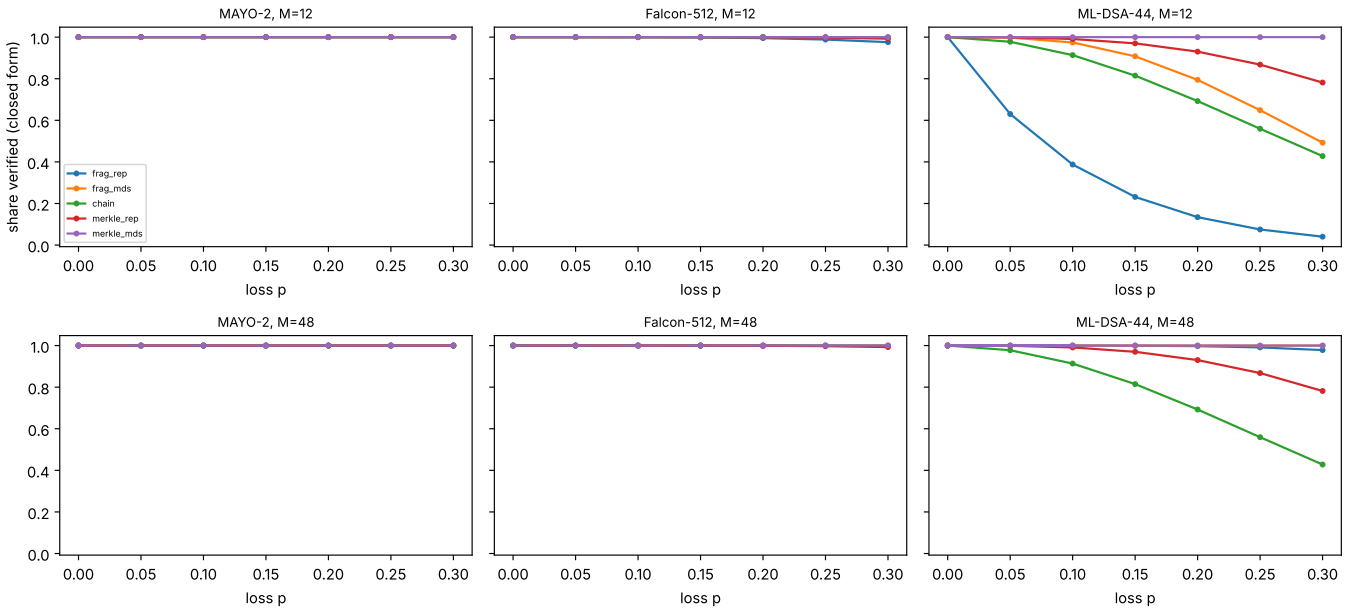


Fig. 2. Share of messages verified (closed form, i.i.d. slot loss) at equal channel use, $M = 12$ (top) and $M = 48$ (bottom) slots per message, SIB1 slot model. Assumed chain and batch parameters (Sec. III). Provisional.

RESULT) of [2]; the 71-block key figure (TODO-RESULT) and 67 blocks (we get 68) of [3]; the ZK compression of [10] (TODO-RESULT); the DNS transfer sizes of [9] (TODO-RESULT); and the tables of [12] (TODO-RESULT).

- **Stateful schemes.** LMS/XMSS sizes are from RFC tables or derived from field lengths, without timings or levels.

VI. Conclusion

Under budgets taken from each channel's sources, only UOV, SNOVA and MAYO-2 carry a full PQ signature in one Remote ID, SIB1 or OSNMA message. Falcon needs the larger C-V2X J3161 or DNS ceilings, and certificates fit only where keys travel out of band. A PQ-anchored chain makes every scheme's per-message cost fit, but it moves the cost into anchor and join time. At equal channel use, it neither dominates fragmentation nor batch signing, and it loses for ML-DSA-44. These are provisional x86 accounting results without security analysis. The next steps are publication runs, MCU timings, measured loss traces, and a per-mode security comparison.

Artifact

Scripts that regenerate every number from raw CSVs: [TODO: artifact URL].

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